

Just Kitchen: Using data analytics to deliver quality cuisines quickly to every doorstep

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[About Just Kitchen](#)

To offer fast food from its virtual restaurants, Just Kitchen runs a stop restaurant management system using data analytics with Google Cloud to optimize its operations and reduce costs.



About Just Kitchen

Established in 2019, Just Kitchen offers food delivery services from its virtual kitchens in Taiwan, Hong Kong, Malaysia, Thailand, the Philippines and Japan. Leveraging data analytics to shorten food delivery time and select sought-after cuisines, Just Kitchen currently processes more than 200,000 orders every month with its own delivery-only brands and dozens of restaurant partners.

Industries: Food & Beverage

Google Cloud results

- Supports exponential data growth with Cloud SQL
- Helps save 30% of operational costs with custom VMs on Compute Engine
- Guarantees nearly zero downtime despite 10X traffic surges through Cloud Load Balancing
- Enables fast content delivery for smooth food order processing

Reduced food processing time by

Location: Taiwan

Compute Engine



About GAIA

Founded in 2015 in Taiwan, GAIA offers a wide range of one-stop cloud integration consulting services covering cloud computing, information security defense and network acceleration.

Being a reseller for more than 10 international tech brands in Taiwan, GAIA has helped 600+ enterprises implement cloud transformation. In 2019, GAIA became a Google Cloud partner.

Google Cloud (<https://cloud.google.com/>) offers a

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

Cloud SQL (<https://cloud.google.com/sql/>) and

Compute Engine

(<https://cloud.google.com/compute/>)

Google Maps Platform

(<https://mapsplatform.google.com/>)

with Cloud CDN

Food delivery is gaining popularity around the world, particularly in densely populated places like Taiwan. During COVID-19, the Taiwanese food delivery market reached

(<https://tw.news.yahoo.com/%E5%A4%96%E9%80%81%E9%A4%90%E9%A3%B2%E6%88%90%E9%95%B7%E7%8E%87%E8%BF%913%E5%80%8D%E8%B6%85%E8%BB%8A%E5%9C%8B%E5%85%A7%E6%89%80%E6%9C%89%E8%A1%8C%E6%A5%AD-091430470.html>)

a 293.78% annual growth. While enjoying the convenience of the service, consumers also want their food to be delivered as fresh and warm as being served in restaurants, and constantly look for more cuisine options.

Just Kitchen (<https://en.justkitchen.com/>) is dedicated to fulfilling consumers' demand for the quality and variety of delivery food. Founded in 2019, it offers quick food delivery services and a wide range of cuisines by establishing virtual restaurants housing various brands, and setting up a streamlined order preparation process that ensures food is ready in the shortest time possible and delivered from the restaurant nearest to the customers. Starting its services in Taiwan, Just Kitchen has expanded its services to Hong Kong, Malaysia, Thailand, the Philippines and Japan, and currently processes more than 200,000 orders every month with its own delivery-only brands and dozens of restaurant partners.

To keep optimizing its operations to deliver higher service quality and better understand its customers' preferences, Just Kitchen built a one-stop platform called JKOS that aggregates all its management systems from supply chain, enterprise resource planning, to point of sale and online ordering. This way, it can enhance management efficiency across all its virtual restaurants while keeping its business data in one place to facilitate data analytics for decision-making.

For higher scalability to accommodate future growth, Just Kitchen decided to run JKOS in a public cloud environment and chose to use Google Cloud (<https://cloud.google.com/>). This is because of its data analytics tools that are highly compatible with third-party software, as well as cost-effective infrastructure.

"To offer excellent food delivery services, we rely heavily on data analysis to improve our operations and minimize food preparation time. That's why having powerful data analytics tools is crucial for us, and Google Cloud meets our needs," says Liu.

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Enhancing food delivery service quality through data analytics

With the technical advice provided by Google Cloud partner GAIA (<https://www.gaia.net/en>) on architecture design and cloud product selection, Just Kitchen launched its first service on Google Cloud after six months of development and completed the whole JKOS platform in May 2022. The organization now uses Cloud SQL (<https://cloud.google.com/sql>) to store all its sales, customer and supply chain data, and leverages BigQuery (<https://cloud.google.com/bigquery>)

in combination with third-party software for data analytics to improve its operations and understand its customers' preferences better.

To shorten the food delivery time, Just Kitchen

employs Google Maps Platform

(<https://mapsplatform.google.com/>) to calculate estimated delivery time and identify the virtual kitchen that's nearest to the customers to prepare the orders. The address and delivery data on Google Maps Platform are regularly scraped and seamlessly imported into Cloud SQL for further processing.

According to Liu, having all its data stored and processed in Google Cloud means higher data visibility, which enables the Just Kitchen team to pinpoint the bottlenecks for service improvement more easily. For example, with the query feature of Cloud SQL, Just Kitchen can quickly gather the data around the time required for each order processing step, which are then sent for in-depth analysis to see which step can be optimized. As a result, it successfully shortened the preparation time for each order from 10 to eight minutes.

"Being able to process all our business data in one place through Cloud SQL and BigQuery has helped us greatly improve our food delivery services and operational efficiency," says Liu. "On top of that, the high scalability of Cloud SQL enables us to manage 10 to 20X more data when we launch a new service. We can effortlessly execute our data-driven strategies no matter how big our business grows."

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Reducing operational costs with custom VMs on Compute Engine

Just Kitchen employs virtual machines (VMs) on Compute Engine (<https://cloud.google.com/compute>) as servers of the JKOS platform. Liu says that being able to customize VM specifications has been beneficial for the organization, particularly at the beginning stages of platform development. This is because they did not have to purchase excess computing resources, therefore allowing more budget for product development. By using the most appropriate combination of CPUs and memory for its different services, Just Kitchen is able to save 30% of operational costs.

"As a startup, we want to use our budget smartly and invest as much as possible in service improvement. The cost-effective infrastructure offered by Compute Engine enables us to smoothly run our platform without wasting any computing resources," he adds.



Ensuring excellent user experience with Cloud Load Balancing and Cloud CDN

Just Kitchen's business has grown rapidly since its launch. When the Taiwanese government announced restrictions on indoor dining at restaurants during COVID-19, the amount of orders that Just Kitchen received on its ordering system increased by 10X the next day. By leveraging Cloud Load Balancing

(<https://cloud.google.com/load-balancing>) to distribute incoming traffic across multiple servers, Just Kitchen was able to realize nearly zero downtime despite any traffic surges, including unpredictable ones that occur during bad weather.

Just Kitchen also uses Cloud CDN

(<https://cloud.google.com/cdn>) to speed up online orders. Before adopting Cloud CDN, the Just Kitchen team conducted a comparison test and found that Cloud CDN offers faster and more stable connectivity than other similar offerings. The extensive global network of Google Cloud also meant that the company's fast content delivery speed remained consistent in different countries when it expanded the business beyond Taiwan, with smooth cross-region data transfers.

"Minimizing service downtime and processing online orders as fast as possible are crucial for us to provide a great user experience and retain our customers," says Liu. "Cloud Load Balancing and Cloud CDN enables us to guarantee stable and quick

online services during unexpected traffic surges and easily expand to different countries."



Streamlining data pipelines to fully unleash the power of data

To enhance the productivity of its employees, Just Kitchen is in the process of adopting Google Workspace (<https://workspace.google.com/>) to facilitate internal collaboration. It also plans to move its business intelligence workloads from third-party platforms to Looker Studio (<https://lookerstudio.google.com/overview>) and Looker (<https://cloud.google.com/looker>) to further streamline its data pipelines and prevent data loss during data transfers between different systems. Having built a comprehensive data ecosystem on Google Cloud, the organization is ready to unleash the power of its business data. For example, it plans to predict food material demand based on past order records, so that it can manage stockage accordingly to reduce waste.

Liu says, "The high-performance cloud infrastructure and data analytics tools of Google Cloud have made it possible for us to optimize our virtual kitchen services and expand our business in a short period of time. We're confident that we can achieve more by taking full advantage of the Google Cloud products in the future."

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